

Power BI — Setup & Unblock Guide

Current state

Layer	Status
Dashboard embedding (/dashboard)	Working
AI tools: diagnose, discover datasets, report pages, visuals	Working
AI tools: DAX queries, schema, visual export	Blocked — needs Azure config below
User access to "PRISM Production" workspace	Blocked — needs Pro license

Part 1 — Assign a Power BI Pro license to your account

Your user account sees "*Upgrade to a paid Power BI license*" because a Pro or PPU license is required to access shared workspaces.

Steps (Microsoft 365 Admin Center)

1. Go to admin.microsoft.com → **Billing** → **Licenses**
2. Assign a **Power BI Pro** (or **Premium Per User**) license to your user account
3. Wait up to 10 minutes for propagation
4. Sign out and back in at app.powerbi.com
5. You can now open the "PRISM Production" workspace

Cost: ~\$10/user/month for Pro, ~\$20/user/month for PPU

Part 2 — Enable service principal API access (unblocks 3 AI tools)

The service principal `98a8da1c-192c-4261-98c6-6cb29482c1f3` is already created. It needs two things:

2a. Grant tenant-level API permissions (Azure Portal)

1. Go to portal.azure.com → **App registrations** → find the app `98a8da1c-...`
2. **API permissions** → **Add a permission** → **Power BI Service**
3. Add **Application permission**: `Dataset.Read.All`
4. Click **Grant admin consent** (requires Global Admin or Privileged Role Admin)
5. While there, also add `Report.Read.All` (for visual export polling)

2b. Enable service principals in Power BI Admin Portal

1. Go to admin.powerbi.com → **Tenant settings**
2. Find **"Allow service principals to use Power BI APIs"** → **Enabled**
 - Apply to: the security group containing your service principal, or "*The entire organization*"
3. Find **"Allow service principals to use read-only admin APIs"** → **Enabled**

2c. Assign Fabric capacity to the workspace

This is the root cause of the 403 on dataset/report read APIs.

1. In **Azure Portal** → **Fabric capacities**

2. Create or locate a Fabric capacity (minimum **F2** SKU)
 - F2: ~~\$0.10/hour~~ (\$130/month, can pause when not in use)
3. In app.powerbi.com → Workspace "**PRISM Production**" → **Settings** → **Premium**
4. Under "*License mode*" select **Fabric capacity** and choose the F2 capacity you created

2d. Add service principal to the workspace

1. In app.powerbi.com → Workspace "**PRISM Production**" → **Manage access**
2. **Add people or groups** → paste the service principal GUID: 98a8da1c-192c-4261-98c6-6cb29482c1f3
3. Assign **Member** or **Contributor** role

Part 3 — Verify everything works

After completing all steps above, run from this project root:

```
node -e "
const env=Object.fromEntries(require('fs').readFileSync('.env','utf-8').split('\n').map(l=>l.trim()).filter(l=>l&&!l.startsWith('#')).map(l=>{const eq=l.indexOf('=');return eq>0?[l.slice(0,eq),l.slice(eq+1)]:[l,'']}));
(async()=>{
  const t=await
  fetch('https://login.microsoftonline.com/'+env.POWERBI_TENANT_ID+'/oauth2/v2.0/token',
    {method:'POST',headers:{'Content-Type':'application/x-www-form-urlencoded'},body:new
    URLSearchParams({client_id:env.POWERBI_CLIENT_ID,client_secret:env.POWERBI_CLIENT_SECRET,grant_type:'client_credentials'})
    const d=await t.json();const b=d.token_type+' '+d.access_token;
    // Test dataset schema (was blocked)
    const s=await
    fetch('https://api.powerbi.com/v1.0/myorg/groups/'+env.POWERBI_WORKSPACE_ID+'/datasets/'+env.POWERBI_DATASET_ID,
    {headers:{Authorization:b}});
    console.log('Schema:',s.status===200?'WORKING':'FAILED HTTP '+s.status);
    // Test DAX
    const q=await
    fetch('https://api.powerbi.com/v1.0/myorg/groups/'+env.POWERBI_WORKSPACE_ID+'/datasets/'+env.POWERBI_DATASET_ID,
    {method:'POST',headers:{Authorization:b,'Content-Type':'application/json'},body:JSON.stringify({queries:[{query:'EVALUATE (VALUES (Table1))'},serializerSettings:{includeNulls:true}})});
    console.log('DAX query:',q.status===200?'WORKING':'FAILED HTTP '+q.status);
})();
"
```

Expected output:

```
Schema: WORKING
DAX query: WORKING
```

Summary — who does what

Step	Who	System
Assign Pro license to your account	M365 Admin	admin.microsoft.com

Grant API permissions + admin consent	Azure Admin	portal.azure.com
Enable service principal tenant settings	Power BI Admin	admin.powerbi.com
Assign Fabric F2 to workspace	Power BI Admin	app.powerbi.com
Add service principal to workspace	Workspace Owner	app.powerbi.com